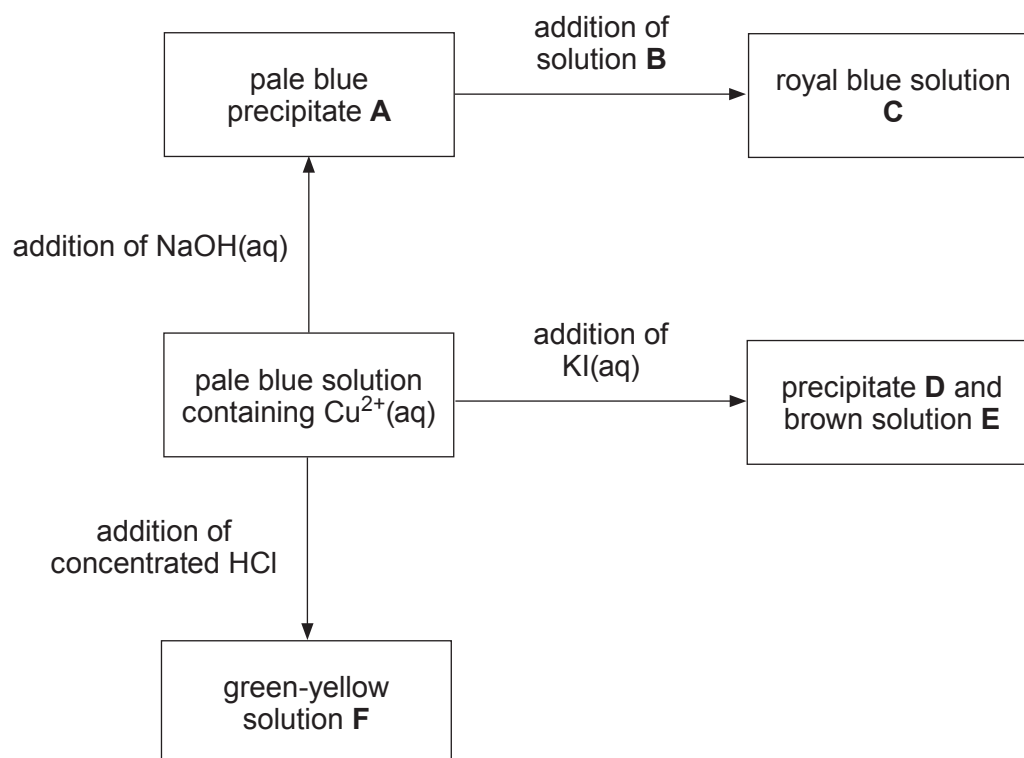


12. (a) The diagram below shows some reactions of copper(II) ions in solution.



Identify the species present in each of the following.

[3]

A

B

C

D

E

F



- (b) Vanadium exhibits the properties of a typical transition element, such as having variable oxidation states and forming coloured complexes. Explain why transition elements have variable oxidation states. [1]

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- (c) Blue oxovanadium(IV) cations, VO_2^{2+} , can be oxidised to yellow oxovanadium(V) ions, VO_2^+ , by bromate(V) ions, BrO_3^- .

	Standard electrode potential, E^θ / V
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$	+1.00
$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.09
$\text{BrO}_3^- + 6\text{H}^+ + 6\text{e}^- \rightleftharpoons \text{Br}^- + 3\text{H}_2\text{O}$	+1.44

Use these standard electrode potentials to identify the main bromine-containing product formed when excess bromate(V) ions are added to a solution containing VO_2^{2+} ions. Give reasons for your answer. [3]

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- (d) The effect of the concentration of H^+ ions on the rate of the reaction in part (c) was studied using the same concentrations and volumes of all other reactants. The following data were collected.

pH	Rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1.0	3.287×10^{-3}
1.4	1.308×10^{-3}
1.8	5.193×10^{-4}
2.2	2.071×10^{-4}

Find the order of the reaction with respect to H^+ . You **must** show your working.

[3]

Order with respect to H^+ =



- (e) The value of the rate constant, k , for this reaction at two temperatures is given below.

Temperature / K	Rate constant (unit not shown)
288	0.0761
300	0.2873

The activation energy for this reaction is $79.333 \text{ kJ mol}^{-1}$ and the Arrhenius equation is:

$$k = Ae^{-\left(\frac{E_a}{RT}\right)}$$

Calculate the value of the rate constant, k , at a temperature of 295 K.

[4]

k at 295 K =



- (f) VO^{2+} ions may also be oxidised to VO_2^+ ions using coloured Ce^{4+} , which is itself reduced to colourless Ce^{3+} . A titration using this reaction does not usually include an indicator and colorimetry is sometimes used.

State why an indicator may not be useful in this reaction and explain how colorimetry can be used to find the end point of the titration. [3]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2

**GCE A LEVEL**

1410U30-1



S18-1410U30-1

CHEMISTRY – A2 unit 3**Physical and Inorganic Chemistry**

TUESDAY, 5 JUNE 2018 – AFTERNOON

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
Section A 1. to 7.	10	
Section B 8.	14	
9.	18	
10.	12	
11.	18	
12.	8	
Total	80	

1410U301
01**ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.12(b)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



JUN181410U30101

12. A student was provided with five ionic solids containing familiar cations and anions. Each solid contains a different **cation**. The solids were labelled **A**, **B**, **C**, **D** and **E**.

The student attempted to dissolve all five solids in water. She then attempted to dissolve those that did not dissolve in water in a stoichiometric amount of acid.

The results are given below.

Solid	Addition of water	Addition of dilute sulfuric acid to solid	Addition of dilute nitric acid to solid
A	dissolves giving colourless solution 1		
B	dissolves giving colourless solution 2		
C	does not dissolve	pale blue solution 3 forms upon warming with the acid	
D	does not dissolve	effervescence and solution 4 forms	
E	does not dissolve	some effervescence but the solid does not dissolve	effervescence and solution 5 forms

Pairs of the solutions formed were mixed and the observations recorded.

	Solution 5	Solution 4	Solution 3	Solution 2
Solution 1 (formed by dissolving solid A)	no visible change	white precipitate	white precipitate	bright yellow precipitate
Solution 2 (formed by dissolving solid B)	no visible change	no visible change	brown solution with a white solid	
Solution 3 (formed by reacting solid C with acid)	thick white precipitate	no visible change		
Solution 4 (formed by reacting solid D with acid)	thick white precipitate			

Flame tests

Flame tests carried out on solutions **1**, **2**, **3**, **4** and **5** gave no colour with one solution, an apple-green flame with another and a lilac flame with a third. The student noted unfamiliar flame test colours for the other two solutions.



Examiner
only

- (a) Suggest which anion is present in substances **D** and **E**. Give a reason for your answer. [2]

- (b) Use **all** the information provided to suggest identities for compounds **A-E**.

Explain clearly how you identified compounds **A** and **C**.

[6 QER]

A _____

B

C

D

E

Explanation

END OF PAPER

8



Surname	Centre Number	Candidate Number
Other Names		2

**GCE A LEVEL**

1410U30-1



S19-1410U30-1

TUESDAY, 4 JUNE 2019 – AFTERNOON**CHEMISTRY – A2 unit 3****Physical and Inorganic Chemistry**

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
Section A 1. to 8.	10	
Section B 9.	15	
10.	15	
11.	12	
12.	15	
13.	13	
Total	80	

1410U301
01**ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11(b)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



JUN191410U30101

12. (a) A student planned to distinguish between the following eight compounds.

magnesium hydroxide

magnesium carbonate

iron(II) hydroxide

iron(II) carbonate

chromium(III) hydroxide

chromium(III) carbonate

lead(II) hydroxide

lead(II) carbonate

He used the following method.

Step 1 Add dilute acid until all the solid has disappeared. Record any effervescence.

Step 2 Add 1 cm³ of sodium hydroxide solution to each solution formed in step 1. Record any precipitate observed.

(i) State which compound(s) would give effervescence and why they do so. [2]

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(ii) The student plans to use dilute hydrochloric acid in step 1. His teacher tells him that this is **not** the correct acid to use.

Explain why hydrochloric acid should not be used and suggest an appropriate acid to use in its place. [2]

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- (iii) Give the colours of the precipitates formed when sodium hydroxide solution is added to solutions containing the following ions. [2]

$\text{Mg}^{2+}(\text{aq})$

$\text{Fe}^{2+}(\text{aq})$

$\text{Cr}^{3+}(\text{aq})$

$\text{Pb}^{2+}(\text{aq})$

- (iv) The method given opposite is incomplete.

Suggest an additional step that would allow the remaining solutions formed in step 2 to be identified. Give the expected observations. State the property of the metals that allows them to be distinguished in this way. [3]

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- (b) Many rocks that contain carbonate anions also contain a mixture of cations. Atomic absorption spectroscopy can be used to find the ratio of the amounts of different cations present.

Analysis of an acid solution formed from the carbonate mineral huntite shows that it contains 91 ng cm^{-3} of magnesium ions and 50 ng cm^{-3} of calcium ions. These are the only two metal cations present.

- (i) Find the formula of the mineral huntite. [3]

Formula

- (ii) The initial solution was prepared using $220 \mu\text{g}$ of huntite. Calculate the volume of aqueous acid used to form the solution for analysis. [3]

Volume = dm^3



12. The formula of an unknown mineral is $\text{Mg}_a\text{X}_b(\text{CO}_3)_c(\text{OH})_d \cdot e\text{H}_2\text{O}$.

A student performs a series of experiments to calculate the values **a**, **b**, **c**, **d** and **e** and name element X.

Experiment Number	Experiment	Result
1	Heating a sample of 0.0200 mol of the mineral to dryness.	The sample loses 1.44 g of mass.
2	Addition of 0.0200 mol of the mineral to excess hydrochloric acid.	A volume of 490 cm^3 of carbon dioxide gas is released at 298 K and 1 atm pressure. The remaining solution has a green colour.
3	Addition of excess aqueous sodium hydroxide gradually to the solution produced in experiment 2.	A precipitate forms that shows a mixture of white and grey-green colours. Addition of excess sodium hydroxide leaves a white precipitate in a green solution.
4	Addition of a sample of 1.00×10^{-3} mol of the mineral to 25.0 cm^3 of HCl of concentration 1.00 mol dm^{-3} . Titration of the remaining acid against standard aqueous sodium hydroxide of concentration 0.500 mol dm^{-3} .	A volume of 14.00 cm^3 of the aqueous sodium hydroxide is required for complete reaction.

(a) Find the value of **e**, the number of water molecules in each formula unit. [2]

e =

(b) Find the value of **c**, the number of carbonate ions in each formula unit. [2]

c =

